SUMIRAN SHAH

sumiranshah@gmail.com | (650) 237-9997 | Santa Clara, CA | https://github.com/sumiran

SUMMARY

Org Tech Lead & Software Engineer with 12+ years of experience architecting mission-critical distributed systems, automated control planes, and large-scale compute/accelerator optimization infrastructure at Google Cloud, YouTube, and Microsoft. Proven track record of defining multi-year technical roadmaps, incubating 0-to-1 platform services through organizational handover, and uniting mathematical optimization (MIP) with production agentic AI governing multi-billion-dollar CapEx decisions.

CORE COMPETENCIES

- **Languages:** Go, C++, Java, Python, TypeScript, JavaScript, SQL, C#
- **Distributed Systems & Infrastructure:** High-Throughput Microservices, Automated Control Planes, Distributed Compute & Pipeline Orchestration, Multi-Dimensional Resource Allocation
- **AI, Optimization & Decision Systems:** Production Agentic Systems, MCP, RAG, Automated Evaluation Loops & Deterministic Guardrails, Gemini API, Mixed-Integer Programming (MIP) Solvers, Large-Scale Simulation Modeling

---

WORK EXPERIENCE

Google - Google Cloud Platform | Sunnyvale, CA | **Tech Lead, Cloud Capacity Planning & Analytics** Aug *2022 - Present*

- **Supply-Demand Matching Architecture:** Lead the Supply-Demand Matching team and define the multi-year architectural roadmap across partner infrastructure, planner, and operations teams. Architected MIP-solver simulation and optimization infrastructure with fault-tolerant ingestion and state orchestration pipelines driving **multi-billion-dollar CapEx decisions**, balancing power, hardware, networking, and virtualization constraints while minimizing compute/memory stranding across GPU/TPU allocations.
- **Agentic AI & Solver Explainability:** Engineered a production agentic diagnostics system utilizing the Gemini API, RAG, and a custom MCP bridge to connect MIP solver outputs with LLM reasoning loops. Built continuous automated evaluation loops and deterministic verification guardrails to benchmark reasoning accuracy and prevent hallucinations, reducing planner bottleneck triage time by **3-4x**.

Senior Software Engineer, Real-Time Capacity Provisioning & Risk Feb 2019 - Aug 2022

- **Real-Time Capacity Engine & Risk Backtesting**: Designed and implemented an automated, high-availability capacity evaluation and fraud-risk control plane, slashing customer request turnaround from **3 business days to 5 minutes**. Built a historical backtesting framework to quantify risk-signal accuracy and establish data-driven policy guardrails.
- **Team Incubation & Operational Handover:** Defined the service's technical roadmap and led the onboarding and knowledge transfer for a newly hired 8-engineer team and Engineering Manager, executing a seamless handover of all production services and tooling.

Google, YouTube | San Bruno, CA | **Software Engineer, YouTube Living Room** Jan 2015 - Feb 2019

- **Cross-Platform Framework Architecture:** Architected and standardized core frontend framework components powering YouTube Living Room applications across Chromecast, Smart TVs, and Game Consoles, supporting a global footprint of **100M+ monthly active TV screens** and **250M+ hours of daily watch time**.
- **Platform Security & Bot Mitigation**: Led the integration of reCAPTCHA into resource-constrained Living Room clients to neutralize view-botting attacks. Proactively identified and patched multiple critical XSS vulnerabilities across youtube.com.

ZocDoc | New York, NY | **Software Engineer Intern, Sales Data Team** Jun 2014 - Aug 2014

- Designed and implemented a high-performance C#/REST caching framework for multi-million-row Salesforce Bulk Queries, reducing query latency from **50 minutes to under 3 minutes** and unlocking new salesperson workflows.

Microsoft | Hyderabad, India | **Software Development Engineer** & **SDE Intern** *Jul 2012 - Jul 2013* & Jun 2011 - Aug 2011

- Built an automated C# transpiler for the Enterprise Service Bus team to migrate legacy document maps from Gentran to BizTalk, eliminating **90%** of manual engineering effort across 600+ maps.

EDUCATION

Columbia University, New York, NY | **Master of Science in Computer Science** (GPA: 3.71 / 4.0) | *Sept 2013 - Dec 2014*

University of Mumbai, Mumbai, India | **Bachelor of Engineering in Computer Science** | *Aug 2008 - June 2012*